Calculation of Reflection and Transmission Coefficients in SCUFF-TRANSMISSION

Homer Reid

May 9, 2015

Contents

1 The Setup **2**

2 Scattering coefficients from surface currents **4**
2.1 Computation of $\widetilde{\mathbf{b}}(\mathbf{q})$ 6

1 The Setup

SCUFF-TRANSMISSION considers geometries with 2D periodicity, i.e. the structure consists of a unit-cell geometry of finite extent in the z direction that is infinitely periodically replicated in both the x and y directions. The structure is illuminated either from below (the default) or from above by a plane wave with propagation vector $\mathbf{k}$ confined to the xz plane.

Working at angular frequency ω, let the free-space wavelength be $k_0 = \frac{\omega}{c}$, and let the relative permittivity and permeability of the lowermost and uppermost regions in the geometry be $\epsilon_{\mathrm{L,U}}$ and $\mu_{\mathrm{L,U}}$. The wavenumber, refractive index, and relative wave impedance in the uppermost and lowermost regions are

$$k_{\mathrm{L}} = n_{\mathrm{L}} \cdot k_0, \qquad n_{\mathrm{L}} \equiv \sqrt{\epsilon_{\mathrm{L}}\mu_{\mathrm{L}}}, \qquad Z_{\mathrm{L}} = \sqrt{\frac{\mu_{\mathrm{L}}}{\epsilon_{\mathrm{L}}}}$$

$$k_{\mathrm{U}} = n_{\mathrm{U}} \cdot k_0, \qquad n_{\mathrm{U}} \equiv \sqrt{\epsilon_{\mathrm{U}}\mu_{\mathrm{U}}}, \qquad Z_{\mathrm{U}} = \sqrt{\frac{\mu_{\mathrm{U}}}{\epsilon_{\mathrm{U}}}}.$$

I will refer to region from which the planewave originates (either the uppermost or lowermost homogeneous region in the SCUFF-EM geometry) as the "incident" region. The region into which the planewave eventually emanates is the "transmitted" region. I will use the sub/superscripts I, T to denote these quantities; thus the wavenumber and relative wave impedance in the incident and transmitted regions are

$$\Big\{k_{\mathrm{I}}, Z_{\mathrm{I}}, k_{\mathrm{T}}, Z_{\mathrm{T}}\Big\} = \begin{cases} \big\{k_{\mathrm{L}}, Z_{\mathrm{L}}, k_{\mathrm{U}}, Z_{\mathrm{U}}\big\}, & \text{wave incident from below} \\ \big\{k_{\mathrm{U}}, Z_{\mathrm{U}}, k_{\mathrm{L}}, Z_{\mathrm{L}}\big\}, & \text{wave incident from above} \end{cases}$$

In what follows, I will use the symbols $\mathbf{k}^{\mathrm{I}}, \mathbf{k}^{\mathrm{R}}, \mathbf{k}^{\mathrm{T}}$ respectively to denote the propagation vectors of the incident, reflected, and transmitted waves. I will take these vectors always to live in the xz plane (i.e. $\mathbf{k}$ has no y component, $k_y = 0$). I will let θ_{I} and θ_{T} be the angles of incidence and transmission (the angles between the incident and transmitted wavevector and the z axis).

$$\text{wave incident from below:} \qquad \mathbf{k}^{\mathrm{I}} = k_{\mathrm{L}}\Big[\sin\theta_{\mathrm{I}}\,\hat{\boldsymbol{x}} + \cos\theta_{\mathrm{I}}\,\hat{\boldsymbol{z}}\Big]$$

$$\text{wave incident from above:} \qquad \mathbf{k}^{\mathrm{I}} = k_{\mathrm{U}}\Big[\sin\theta_{\mathrm{I}}\,\hat{\boldsymbol{x}} - \cos\theta_{\mathrm{I}}\,\hat{\boldsymbol{z}}\Big]$$

The reflected wavevector is

$$\text{wave incident from below:} \qquad \mathbf{k}^{\mathrm{R}} = k_{\mathrm{L}}\Big[\sin\theta_{\mathrm{I}}\,\hat{\boldsymbol{x}} - \cos\theta_{\mathrm{I}}\,\hat{\boldsymbol{z}}\Big]$$

$$\text{wave incident from above:} \qquad \mathbf{k}^{\mathrm{R}} = k_{\mathrm{L}}\Big[\sin\theta_{\mathrm{I}}\,\hat{\boldsymbol{x}} + \cos\theta_{\mathrm{I}}\,\hat{\boldsymbol{z}}\Big]$$

The transmitted wavevector is

$$\text{wave incident from below:} \qquad \mathbf{k}^{\mathrm{T}} = k_{\mathrm{U}}\Big[\sin\theta_{\mathrm{T}}\,\hat{\boldsymbol{x}} + \cos\theta_{\mathrm{T}}\,\hat{\boldsymbol{z}}\Big]$$

$$\text{wave incident from above:} \qquad \mathbf{k}^{\mathrm{T}} = k_{\mathrm{L}}\Big[\sin\theta_{\mathrm{T}}\,\hat{\boldsymbol{x}} - \cos\theta_{\mathrm{T}}\,\hat{\boldsymbol{z}}\Big]$$

The incident and transmitted angles are related by

$$n_{\rm I}\sin\theta_{\rm I} = n_{\rm T}\sin\theta_{\rm T}.$$

For a general vector $\mathbf{v}$, I will define $\widehat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$ to be a unit vector in the direction of $\mathbf{v}$.

Definition of scattering coefficients

The incident, reflected, and transmitted fields may be written in the form

$$\begin{array}{rclcrcl}
\mathbf{E}^{\rm I}(\mathbf{x}) &=& E_0\boldsymbol{\epsilon}^{\rm I}e^{i\mathbf{k}^{\rm I}\cdot\mathbf{x}} & & \mathbf{H}^{\rm I}(\mathbf{x}) &=& H_0\overline{\boldsymbol{\epsilon}}^{\rm I}e^{i\mathbf{k}^{\rm I}\cdot\mathbf{x}}l \\
\mathbf{E}^{\rm R}(\mathbf{x}) &=& rE_0\boldsymbol{\epsilon}^{\rm R}e^{i\mathbf{k}^{\rm R}\cdot\mathbf{x}} & & \mathbf{H}^{\rm R}(\mathbf{x}) &=& rH_0\overline{\boldsymbol{\epsilon}}^{\rm R}e^{i\mathbf{k}^{\rm R}\cdot\mathbf{x}} \\
\mathbf{E}^{\rm T}(\mathbf{x}) &=& tE_0\boldsymbol{\epsilon}^{\rm T}e^{i\mathbf{k}^{\rm T}\cdot\mathbf{x}} & & \mathbf{H}^{\rm T}(\mathbf{x}) &=& tH_0'\,\overline{\boldsymbol{\epsilon}}^{\rm T},e^{i\mathbf{k}^{\rm T}\cdot\mathbf{x}}
\end{array} \qquad (1)$$

where E_0 is the incident field magnitude, $\boldsymbol{\epsilon}^{\rm I,R,T}$ are **E**-field polarization vectors, $\overline{\boldsymbol{\epsilon}}^{\rm I,R,T}$ are **H**-field polarization vectors, and we have

$$H_0 \equiv \frac{i|\mathbf{k}|E_0}{Z_{\rm I}Z_0}, \qquad H_0' \equiv \frac{i|\mathbf{k}|E_0}{Z_{\rm T}Z_0}, \qquad \overline{\boldsymbol{\epsilon}} \equiv \widehat{\mathbf{k}}\times\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} = -\widehat{\mathbf{k}}\times\overline{\boldsymbol{\epsilon}}.$$

The polarization vectors are given by

$$\begin{array}{lll}
\text{for the TE case:} & \boldsymbol{\epsilon}^{\rm I}_{\rm TE} = \boldsymbol{\epsilon}^{\rm R}_{\rm TE} = \boldsymbol{\epsilon}^{\rm T}_{\rm TE} = \hat{\boldsymbol{y}}, & \overline{\boldsymbol{\epsilon}}^{\rm I,R,T}_{\rm TE} = \widehat{\mathbf{k}^{\rm I,R,T}}\times\hat{\boldsymbol{y}} \\
\text{for the TM case:} & \overline{\boldsymbol{\epsilon}}^{\rm I}_{\rm TM} = \overline{\boldsymbol{\epsilon}}^{\rm R}_{\rm TM} = \overline{\boldsymbol{\epsilon}}^{\rm T}_{\rm TM} = \hat{\boldsymbol{y}} & \boldsymbol{\epsilon}^{\rm I,R,T}_{\rm TM} = -\widehat{\mathbf{k}^{\rm I,R,T}}\times\hat{\boldsymbol{y}}
\end{array}$$

Equations (1) define the reflection and transmission coefficients r and t computed by SCUFF-TRANSMISSION.

2 Scattering coefficients from surface currents

Next we consider an extended structure described by Bloch-periodic boundary conditions, i.e. all fields and currents satisfy

$$\mathbf{Q}(\mathbf{x}+\mathbf{L}) = e^{i\mathbf{k}_{\rm B}\cdot\mathbf{L}}\,\mathbf{Q}(\mathbf{x}) \tag{2}$$

where $\mathbf{Q}$ is a field ($\mathbf{E}$ or $\mathbf{H}$) or a surface current ($\mathbf{K}$ or $\mathbf{N}$) and the Bloch wavevector is[1]

$$\mathbf{k}_{\rm B} = k_{\rm I}\sin\theta_{\rm I}\,\hat{\boldsymbol{x}} = k_{\rm T}\sin\theta_{\rm T}\,\hat{\boldsymbol{x}}.$$

For plane waves like (1), equation (2) actually holds for any arbitrary vector $\mathbf{L}$; for our purposes we will only need to use it for certain special vectors $\mathbf{L}$ determined by the structure of the lattice in our PBC geometry. We will derive expressions for the plane-wave reflection and transmission coefficients in terms of the surface-current distribution in the unit cell of the structure.

Fields from surface currents

On the other hand, the scattered $\mathbf{E}$ fields in the incident and transmitted regions may be obtained in the usual way from the surface-current distributions on the surfaces bounding those regions. For example, at points in the transmitted medium, the scattered (that is, transmitted) $\mathbf{E}$ field is given by

$$\begin{aligned}\mathbf{E}^{\rm T}(\mathbf{x}) &= \oint_{\mathcal{S}_{\rm T}} \Big\{ \mathbf{\Gamma}^{\rm EE;T}(\mathbf{x},\mathbf{x}')\cdot\mathbf{K}(\mathbf{x}') + \mathbf{\Gamma}^{\rm EM;T}(\mathbf{x},\mathbf{x}')\cdot\mathbf{N}(\mathbf{x}') \Big\} d\mathbf{x}' \\ &= ik_{\rm T}\oint_{\mathcal{S}} \Big\{ Z_0 Z_{\rm T}\mathbf{G}(k_{\rm T};\mathbf{x},\mathbf{x}')\cdot\mathbf{K}(\mathbf{x}') + \mathbf{C}(k_{\rm T};\mathbf{x},\mathbf{x}')\cdot\mathbf{N}(\mathbf{x}') \Big\} d\mathbf{x}' \end{aligned} \tag{3}$$

where $\mathcal{S}_{\rm T}$ is the surface bounding the transmitted region and $\mathbf{G},\mathbf{C}$ are the homogeneous dyadic GFs for that region. Using the Bloch periodicity of the surface currents, i.e.

$$\left\{ \begin{array}{c} \mathbf{K}(\mathbf{x}+\mathbf{L}) \\ \mathbf{N}(\mathbf{x}+\mathbf{L}) \end{array} \right\} = e^{i\mathbf{k}_{\rm B}\cdot\mathbf{L}} \left\{ \begin{array}{c} \mathbf{K}(\mathbf{x}) \\ \mathbf{N}(\mathbf{x}) \end{array} \right\}$$

we can restrict the surface integral in (3) to just the lattice unit cell:

$$\mathbf{E}^{\rm T}(\mathbf{x}) = ik_{\rm T}\int_{\rm UC} \Big\{ Z_0 Z_{\rm T}\overline{\mathbf{G}}(k_{\rm T};\mathbf{x},\mathbf{x}')\cdot\mathbf{K}(\mathbf{x}') + \overline{\mathbf{C}}(k_{\rm T};\mathbf{x},\mathbf{x}')\cdot\mathbf{N}(\mathbf{x}') \Big\}\, d\mathbf{x}' \tag{4}$$

where the periodic Green's functions are

$$\left\{ \begin{array}{c} \overline{\mathbf{G}}(\mathbf{x},\mathbf{x}') \\ \overline{\mathbf{C}}(\mathbf{x},\mathbf{x}') \end{array} \right\} \equiv \sum_{\mathbf{L}} e^{i\mathbf{k}_{\rm B}\cdot\mathbf{L}} \left\{ \begin{array}{c} \mathbf{G}(\mathbf{x},\mathbf{x}'+\mathbf{L}) \\ \mathbf{C}(\mathbf{x},\mathbf{x}'+\mathbf{L}) \end{array} \right\} \tag{5}$$

[1] Recall our convention that the propagation vector lives in the xy plane.

I now invoke the following representation of the dyadic Green's functions (Chew, 1995): for $z > z'$,

$$\mathbf{G}(k;\boldsymbol{\rho},z;\boldsymbol{\rho}',z') = \int \frac{d\mathbf{q}}{(2\pi)^2}\,\widetilde{\mathbf{G}}^{\pm}(k;\mathbf{q})e^{i\mathbf{q}\cdot(\boldsymbol{\rho}-\boldsymbol{\rho}')}e^{\pm iq_z(z-z')} \tag{6a}$$

$$\mathbf{C}(k;\boldsymbol{\rho},z;\boldsymbol{\rho}',z') = \int \frac{d\mathbf{q}}{(2\pi)^2}\,\widetilde{\mathbf{C}}^{\pm}(k;\mathbf{q})e^{i\mathbf{q}\cdot(\boldsymbol{\rho}-\boldsymbol{\rho}')}e^{\pm iq_z(z-z')} \tag{6b}$$

where $\mathbf{q} = (q_x, q_y)$ is a two-dimensional Fourier wavevector, $d\mathbf{q} = dq_x dq_y$, $q_z = \sqrt{k^2 - |\mathbf{q}|^2}$, $\pm = \mathrm{sign}(z - z')$, and

$$\widetilde{\mathbf{G}}^{\pm}(k;\mathbf{q}) = \frac{i}{2q_z}\left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{k^2}\begin{pmatrix} q_x^2 & q_xq_y & \pm q_xq_z \\ q_yq_x & q_y^2 & \pm q_yq_z \\ \pm q_zq_x & \pm q_zq_y & q_z^2 \end{pmatrix}\right]$$

$$\widetilde{\mathbf{C}}^{\pm}(k;\mathbf{q}) = \frac{i}{2q_zk}\begin{pmatrix} 0 & \pm q_z & -q_y \\ \mp q_z & 0 & q_x \\ q_y & -q_x & 0 \end{pmatrix}.$$

Inserting (6) into (5), I obtain, for the periodic version of e.g. the $\mathbf{G}$ kernel,

$$\begin{aligned}\overline{\mathbf{G}}(k;\boldsymbol{\rho},z;\boldsymbol{\rho}',z) &= \int \frac{d\mathbf{q}}{(2\pi)^2}\widetilde{\mathbf{G}}^{\pm}(k;\mathbf{q})e^{i\mathbf{q}\cdot(\boldsymbol{\rho}-\boldsymbol{\rho}')}e^{\pm iq_z(z-z')}\underbrace{\sum_{\mathbf{L}} e^{i(\mathbf{k}_\mathrm{B}-\mathbf{q})\cdot\mathbf{L}}}_{\mathcal{V}_\mathrm{BZ}\sum_{\boldsymbol{\Gamma}}\delta(\mathbf{q}-\mathbf{k}-\boldsymbol{\Gamma})} \\ &= \mathcal{V}_\mathrm{UC}^{-1}\sum_{\mathbf{q}=\mathbf{k}_\mathrm{B}+\boldsymbol{\Gamma}}\widetilde{\mathbf{G}}^{\pm}(k;\mathbf{q})e^{i\mathbf{q}\cdot(\boldsymbol{\rho}-\boldsymbol{\rho}')}e^{\pm iq_z(z-z')}\end{aligned}$$

where the sum over $\boldsymbol{\Gamma}$ runs over reciprocal lattice vectors; the prefactor $\mathcal{V}_\mathrm{BZ}$, the volume of the Brillouin zone, is related to the unit-cell volume by $\mathcal{V}_\mathrm{BZ} = 4\pi^2/V_\mathrm{UC}$ for a 2D square lattice. Similarly, we find

$$\overline{\mathbf{C}}(k;\boldsymbol{\rho},z;\boldsymbol{\rho}',z) = \mathcal{V}_\mathrm{UC}^{-1}\sum_{\mathbf{q}=\mathbf{k}_\mathrm{B}+\boldsymbol{\Gamma}}\widetilde{\mathbf{C}}^{\pm}(k;\mathbf{q})e^{i\mathbf{q}\cdot(\boldsymbol{\rho}-\boldsymbol{\rho}')}e^{\pm iq_z(z-z')}.$$

Keeping only the $\boldsymbol{\Gamma} = 0$ term in these sums, the scattered $\mathbf{E}$-fields in the uppermost and lowermost regions are thus

$$\mathbf{E}^\mathrm{upper}(\mathbf{x}) = e^{i(k_\mathrm{Ux}x + k_\mathrm{Uz}z)}\Big[ik_\mathrm{U}Z_0Z_\mathrm{U}\widetilde{\mathbf{G}}^{+}(k_\mathrm{U};\mathbf{k}_\mathrm{B})\widetilde{\mathbf{K}}_\mathrm{U}(\mathbf{k}_\mathrm{B}) + ik_\mathrm{U}\widetilde{\mathbf{C}}^{+}(k_\mathrm{U};\mathbf{k}_\mathrm{B})\widetilde{\mathbf{N}}(\mathbf{k}_\mathrm{B})\Big] \tag{7}$$

$$\mathbf{E}^\mathrm{lower}(\mathbf{x}) = e^{i(k_\mathrm{Lx}x - k_\mathrm{Lz}z)}\Big[ik_\mathrm{L}Z_0Z_\mathrm{U}\widetilde{\mathbf{G}}^{-}(k_\mathrm{L};\mathbf{k}_\mathrm{B})\widetilde{\mathbf{K}}(\mathbf{k}_\mathrm{B}) + ik_\mathrm{L}\widetilde{\mathbf{C}}^{-}(k_\mathrm{L};\mathbf{k}_\mathrm{B})\widetilde{\mathbf{N}}(\mathbf{k}_\mathrm{B})\Big] \tag{8}$$

where e.g. $\widetilde{\mathbf{K}}_\mathrm{U}$ is something like the two-dimensional Fourier transform of the surface currents on the boundary of the uppermost region $\mathcal{R}_\mathrm{U}$:

$$\widetilde{\mathbf{K}}_\mathrm{U}(\mathbf{k}_\mathrm{B}) \equiv \frac{1}{\mathcal{V}_\mathrm{UC}}\int_{\partial\mathcal{R}_\mathrm{U}}\mathbf{K}(\boldsymbol{\rho}',z')e^{-i\mathbf{k}_\mathrm{B}\cdot\boldsymbol{\rho}'-iq_z|z'|}d\mathbf{x}', \qquad q_z^2 = k_\mathrm{U}^2 - |\mathbf{k}_\mathrm{B}|^2.$$

with $\widetilde{\mathbf{K}}_{\rm L}$ and $\widetilde{\mathbf{N}}_{\rm U,L}$ defined similarly.

Comparing this to (1c), we see that the transmission and reflection coefficients for the polarization defined by polarization vector $\boldsymbol{\epsilon}$ are given by

$$\left\{ \begin{array}{c} t \\ r \end{array} \right\} = ik_{\rm U} Z_0 Z_{\rm U} \boldsymbol{\epsilon}^\dagger \widetilde{\mathbf{G}}^+(k_{\rm U}, \mathbf{k}_{\rm B}) \widetilde{\mathbf{K}}_{\rm U}(\mathbf{k}_{\rm B}) + ik_{\rm U} \boldsymbol{\epsilon}^\dagger \widetilde{\mathbf{C}}^+(k_{\rm U}, \mathbf{k}_{\rm B}) \widetilde{\mathbf{N}}(\mathbf{k}_{\rm B}) \qquad (9)$$

$$\left\{ \begin{array}{c} r \\ t \end{array} \right\} = ik_{\rm L} Z_0 Z_{\rm L} \boldsymbol{\epsilon}^\dagger \widetilde{\mathbf{G}}^-(k_{\rm L}, \mathbf{k}_{\rm B}) \widetilde{\mathbf{K}}_{\rm L}(\mathbf{k}_{\rm B}) + ik_{\rm L} \boldsymbol{\epsilon}^\dagger \widetilde{\mathbf{C}}^-(k_{\rm L}, \mathbf{k}_{\rm B}) \widetilde{\mathbf{N}}(\mathbf{k}_{\rm B}) \qquad (10)$$

The expressions on the RHS compute the upper (lower) quantities on the LHS for the case in which the plane wave is incident from below (above).

The $\widetilde{\mathbf{K}}$ and $\widetilde{\mathbf{N}}$ quantities are given by sums of contributions from individual basis functions; for example,

$$\widetilde{\mathbf{K}}_{\rm U}(\mathbf{q}) = \frac{1}{\mathcal{V}_{\rm UC}} \sum_{\mathbf{b}_\alpha \in \partial\mathcal{R}_{\rm U}} k_\alpha \widetilde{\mathbf{b}_\alpha}(\mathbf{q}), \qquad \widetilde{\mathbf{N}}_{\rm U}(\mathbf{q}) = -\frac{Z_0}{\mathcal{V}_{\rm UC}} \sum_{\mathbf{b}_\alpha \in \partial\mathcal{R}_{\rm U}} n_\alpha \widetilde{\mathbf{b}_\alpha}(\mathbf{q})$$

where the sums are over all RWG basis functions that live on surfaces bounding the uppermost medium and $\{k_\alpha, n_\alpha\}$ are the surface-current coefficients obtained as the solution to the SCUFF-EM scattering problem.

2.1 Computation of $\widetilde{\mathbf{b}}(\mathbf{q})$

For RWG functions the quantity $\widetilde{\mathbf{b}}(\mathbf{q})$ may be evaluated in closed form:

$$\begin{aligned}
\widetilde{\mathbf{b}_\alpha}(\mathbf{q}) &\equiv \int_{\sup \mathbf{b}_\alpha} \mathbf{b}_\alpha(\mathbf{x}) e^{-i\mathbf{q}\cdot\mathbf{x}} \, d\mathbf{x} \\
&= \ell_\alpha \int_0^1 du \int_0^u dv \left\{ e^{-i\mathbf{q}\cdot[\mathbf{Q}^+ + u\mathbf{A}^+ + v\mathbf{B}]} \Big(u\mathbf{A}^+ + v\mathbf{B} \Big) \right. \\
&\qquad \left. - e^{-i\mathbf{q}\cdot[\mathbf{Q}^- + u\mathbf{A}^- + v\mathbf{B}]} \Big(u\mathbf{A}^- + v\mathbf{B} \Big) \right\} \\
&= \ell_\alpha \left\{ e^{-i\mathbf{q}\cdot\mathbf{Q}^+} \Big[f_1\big(\mathbf{q}\cdot\mathbf{A}^+, \mathbf{q}\cdot\mathbf{B}\big)\mathbf{A}^+ + f_2\big(\mathbf{q}\cdot\mathbf{A}^+, \mathbf{q}\cdot\mathbf{B}\big)\mathbf{B} \Big] \right. \\
&\qquad \left. - e^{-i\mathbf{q}\cdot\mathbf{Q}^-} \Big[f_1\big(\mathbf{q}\cdot\mathbf{A}^-, \mathbf{q}\cdot\mathbf{B}\big)\mathbf{A}^- + f_2\big(\mathbf{q}\cdot\mathbf{A}^-, \mathbf{q}\cdot\mathbf{B}\big)\mathbf{B} \Big] \right\}
\end{aligned}$$

where

$$\begin{aligned}
f_1(x,y) &= \int_0^1 \int_0^u u e^{-i(ux+vy)} \, dv \, du \\
f_2(x,y) &= \int_0^1 \int_0^u v e^{-i(ux+vy)} \, dv \, du.
\end{aligned}$$